

SIMATS ENGINEERING
CO3-ASSESSMENT TOOL 1
Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Dr.V.Parthipan

Code of Conduct:

I _M.V.PREETHASHRI(192524476)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student M.V.Preethashri

Criteria	Understanding of Concepts 2.5	Experimental Design 3	Use of Tools 2	Analysis of Results 1.5	Documentation 1	Total (10)
Q1	2.5	3	2	1.5	1	10
Q2	2	3	2	1.5	1	9.5
Q3	2.5	2.5	2	1.5	1	9.5
Q4	2.5	2.5	2	1.5	1	9.5

Faculty Signature

Experiments

QUESTIONS:4

Total Marks:40

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, and ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)

The image shows a Wireshark packet capture of a TCP connection setup. The main window displays a list of packets, and a detailed view of packet 721 is shown on the right.

Packet List:

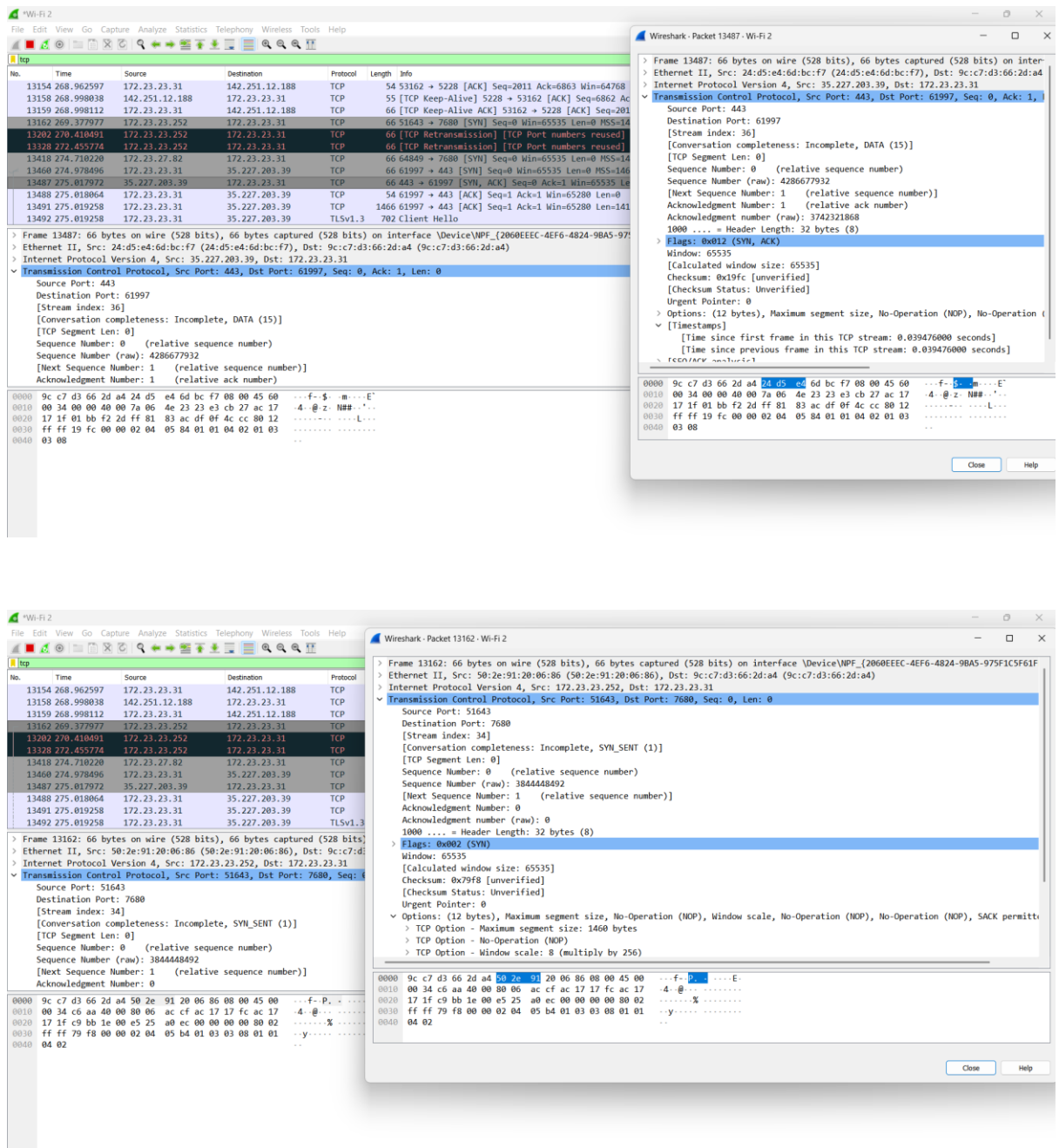
No.	Time	Source	Destination	Protocol	Length	Info
3126	61.761141	142.251.223.243	172.23.23.31	TLSv1.3	672	Application Data, Application Data, Application Data
3127	61.761141	142.251.223.243	172.23.23.31	TCP	60	[TCP Retransmission] Seq=443, Dst Port: 60075
3128	61.761248	172.23.23.31	142.251.223.243	TCP	66	63510 → 443 [ACK] Seq=24091 Ack=7381 Win=65204
3129	61.763336	172.23.23.31	142.251.223.243	TLSv1.3	89	Application Data
3130	61.763400	172.23.23.31	142.251.223.243	TLSv1.3	93	Application Data
3131	61.770520	142.251.223.243	172.23.23.31	TCP	54	443 → 63510 [ACK] Seq=7381 Ack=24126 Win=24725
3132	61.770520	142.251.223.243	172.23.23.31	TCP	54	443 → 63510 [ACK] Seq=7381 Ack=24165 Win=24725
3141	62.163261	172.23.23.84	172.23.23.31	TCP	66	56518 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=65535
3193	63.209702	172.23.23.84	172.23.23.31	TCP	66	[TCP Retransmission] [TCP Port numbers reused]
3288	65.140394	172.23.23.84	172.23.23.31	TCP	66	[TCP Retransmission] [TCP Port numbers reused]
3365	69.225798	172.23.23.84	172.23.23.31	TCP	66	[TCP Retransmission] [TCP Port numbers reused]

Packet 721 Details:

- Frame 721: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface 'Device\NPF_{2060EEEC-4EF6-4824-9BA5-975...}
- Ethernet II, Src: 24:d5:e4:6d:bc:f7 (24:d5:e4:6d:bc:f7), Dst: 9c:c7:d3:66:2d:a4 (9c:c7:d3:66:2d:a4)
- Internet Protocol Version 4, Src: 104.18.39.21, Dst: 172.23.23.31
- Transmission Control Protocol, Src Port: 443, Dst Port: 60075, Seq: 25, Ack: 29, Len: 0

Packet 721 Payload:

```
0000  9c c7 d3 66 2d a4 24 d5 e4 6d bc f7 08 00 45 00  ...f-$-m...E
0010  00 28 24 a4 40 00 3a 06 c9 ce 68 12 27 15 ac 17  ($@:~h'...
0020  17 1f 01 bb ea ab 7b ed 00 16 7b 54 75 24 50 10  ....x{~(Tu$P
0030  00 12 04 82 00 00  ....
```



2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)

Wireshark - Packet 102713 - Wi-Fi 2

Frame 102713: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface \Device\NPF_{2060EEEC-4EF6-4824-9BA5-795...}

Ethernet II, Src: 9c:c7:d3:66:2d:a4 (9c:c7:d3:66:2d:a4), Dst: 24:d5:e4:6d:bc:f7 (24:d5:e4:6d:bc:f7)

Internet Protocol Version 4, Src: 172.23.23.31, Dst: 172.21.100.31

0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 Total Length: 56
 Identification: 0x8147 (33095)
 > Flags: 0x00
 ... 0 0000 0000 0000 = Fragment Offset: 0
 Time to Live: 128
 Protocol: UDP (17)
 Header Checksum: 0xe602 [validation disabled]

0000 24 d5 e4 6d bc f7 9c c7 d3 66 2d a4 08 00 45 00 \$-m-----f---E-
 0010 00 38 81 47 00 00 80 11 e6 02 ac 17 17 1f ac 15 8-G:-----
 0020 64 1f c9 ef 00 35 00 24 5f 48 4d b0 01 00 00 01 d---5\$JM-----
 0030 00 00 00 00 00 00 03 64 6e 73 06 67 6f 6f 67 6c -----d ns googl
 0040 65 00 00 01 00 01 e-----

3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)

Wireshark - Packet 3008 - Wi-Fi 2

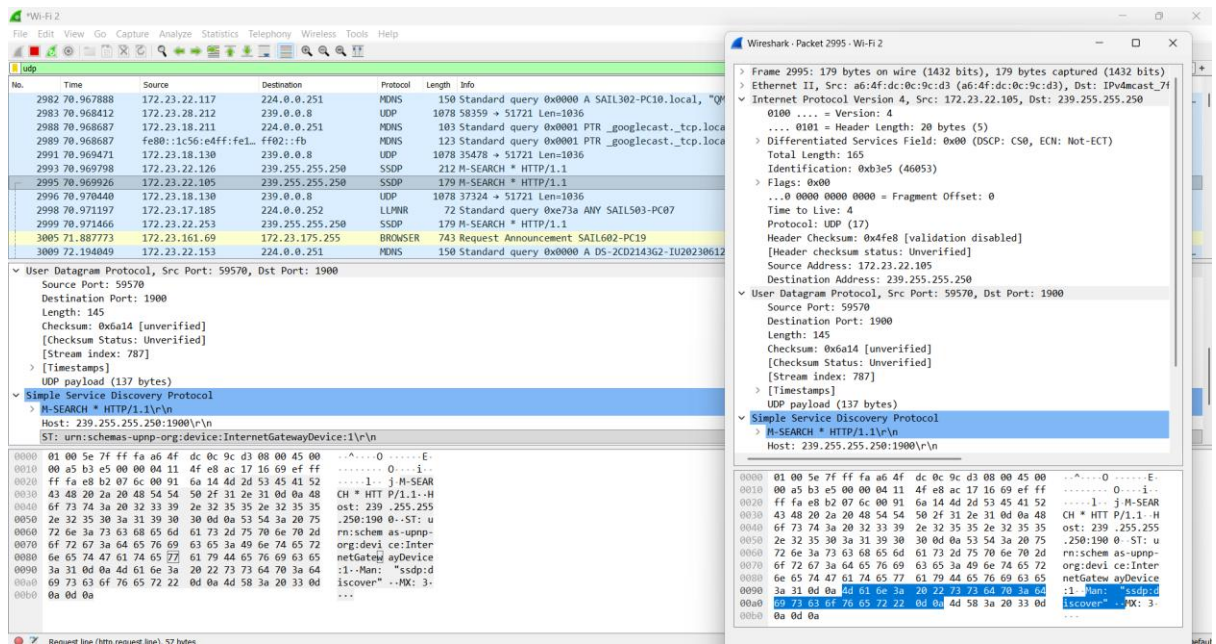
Frame 3008: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{2060EEEC-4EF6-4824-9BA5-795...}

Ethernet II, Src: Liteontec3:58:e5 (d0:39:57:c3:58:e5), Dst: 9c:c7:d3:66:2d:a4 (9c:c7:d3:66:2d:a4)

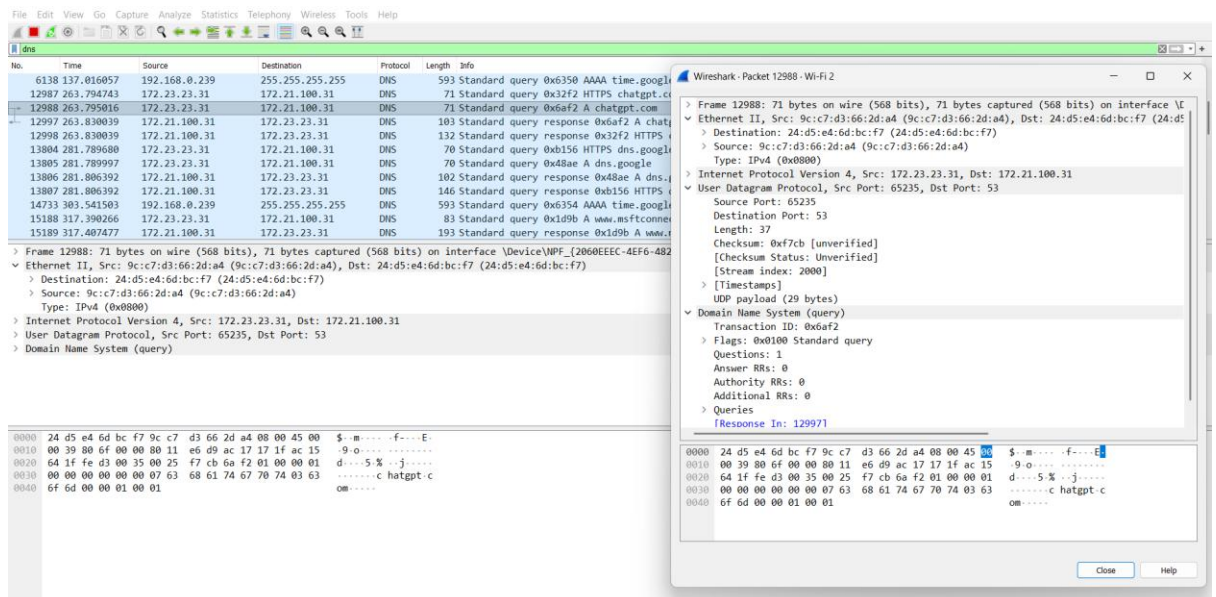
Internet Protocol Version 4, Src: 172.23.21.220, Dst: 172.23.23.31

0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 Total Length: 52
 Identification: 0xb06b (2923)
 > Flags: 0x40, Don't fragment
 ... 0 0000 0000 0000 = Fragment Offset: 0
 Time to Live: 128
 Protocol: TCP (6)
 Header Checksum: 0xa2f [validation disabled]

0000 9c c7 d3 66 2d a4 d0 39 57 c3 58 e5 08 00 45 00 ---f---9 W X---E-
 0010 00 34 0b 6b 40 00 80 06 6a 2f ac 17 15 dc ac 17 4-k@---j/-----
 0020 17 1f e8 3f 1e 00 ba 21 df b4 00 00 00 00 80 02 ...?---I-----
 0030 ff ff 49 d0 00 02 04 05 b4 01 03 08 01 01d I-----
 0040 04 02 ..



4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP.(BL4) (10)



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
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Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation